



Optimization of catalytic pyrolysis process for change of plastic waste into fuel

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ABSTRACT

The utilization of plastic material is huge and consistently developing in the perspective on the favourable position got from the adaptability, minimal effort and tough. Plastic creation has come to at 550 million tons in 2019. The increment being used of plastic items brought about by unexpected development in expectations for everyday comforts had an amazing effect of condition. In this work we convert plastic waste into Fuel by pyrolysis method. Alumina is used as the catalytic material for pyrolysis process. This process is designed to L₉ orthogonal array with provided 3 levels and 3 factors such as catalytic ratio, Temperature and production time are the input process parameters to quantity of fuel produced by pyrolysis process. Optimization of catalytic pyrolysis process parameters are confirmed by analysis of variance.

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1. Introduction

Plastics are hydrocarbons that are produced using oil, and they can be converted back to liquid fuel. Researchers have normally utilized a cycle called pyrolysis to do this, which requires heating the plastics at a high temperature. It is heated to 427 °C (800°F) which changes over the plastic into a liquid. The machine at that point changes the liquid into vaporous state. The gas created is caught and permitted to cool. The vapours condensate when cooled and structure unrefined petroleum. Plastic Pyrolysis is a concoction response. This response includes the sub-atomic breakdown of bigger particles into littler particles within the sight of warmth. Plastic pyrolysis includes exposing plastic to high temperature of 400 to 450 °C, without oxygen. During pyrolysis plastic separates into littler particles of pyrolysis oil, The normally utilized plastics can be characterized into six unique kinds of plastics, for example, Polyethylene terephthalate (PET), High-thickness polyethylene (HDPE), Polyvinyl chloride (PVC), Low-thickness polyethylene (LDPE), Polypropylene (PP) and Polystyrene (PS). We have to provide an appropriate back round examination on differ-

ent innovations and procedures accessible to change of plastic waste into fuel. The variables influencing the synergist pyrolysis cycle, for example, temperature, maintenance time, feed stock structure and utilization of reactant. Reactant pyrolysis of plastic waste has risen with utilization of catalyst. It can possibly convert plastic waste 70%–80% of plastic waste into fluid oil that has comparative attributes to traditional diesel Engine.

2. Literature survey

Literature survey is carried out to know the previous work in the world. They are summarized as follows. Ravi et al. [1] have explained about prepare alternate fuel from the waste plastic, they used reactor, condenser, steam tube and maintained the temperature 350–600°C with atmospheric pressure. As a result found out crude oil which contains three petroleum products gasoline, diesel and petrol with wax as a by product, using fibres, plates, tubes, films, bottles, boxes and different shapes of plastic wastes and end of the process naptha like melted plastic waste which can be used as road construction, the product SO₂ came out as initial stage which is useful to produce H₂SO₄ and preheating. Hayelom [2] focused on recycling of plastic waste into fuels, a review as inputs plastic containers, sheets, hosepipes and focused to use reactor,

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condenser, cracker, receiver tank, storage tank and collection as shipping and found out combustible gas, tars, oils and char. Antony et al. [15] investigated extraction of fuel from waste plastic by pyrolysis with three steps such as cleaning; shredding and distillation process and found the application with the product of pyrolysis process as diesel, petrol and plastic pyrolysis oil. Sanjeev Kumar and Sushant Saklani [11] focused on study the parameters of fuel from waste plastics. The inputs are plastic bottles, waste packets, carry bags and the pyrolysis plant consists of reactor, condenser, catalyst, cooling tube and fuel tank. In this process the waste plastics is changed into carbon smoke and combustible gas which can be used to heat the sample and the important renewable resources used as fuel after refining. The catalyst such CaCl_2 , Al_2O_3 and ZSM-5 and required temperature ranges 350–500 °C for time of optimum conversion takes place exactly one hour and 5 min. Conclusion of the investigation the prepared plastic fuel can be used for diesel electric generators, hot water generators, hot air generators, boilers and this method is better ecologically and economically. Aditya Machiraju et al. [12] experimented on extraction of liquid hydrocarbon fuel from waste plastic and output as char, pyro gas, wax, pyrolytic oil, temperature during entire process is 275 °C and modified pressure cooker of capacity one litre (for polystyrene) without catalyst. The used equipments are bomb calorimeter and saybolt viscometer for finding calorific value and viscosity fire point can be measured by fire point apparatus. Basakaran and Sathish [5] assessed on execution of CI motor with squander plastic oil-diesel mixes as elective fuel with pyrolysis and used reactor, condenser and liquid product collector and inputs are different types of kitchen plastics, obtained output as diesel and plastic oil, results 20% of thermal and mechanical efficiency increased in diesel engine. YasaShukla and Hemant Singh [6] concentrated on plan of practical machine to change over waste plastic into blended oil for local reason in with pyrolysis process in two techniques warm pyrolysis (no catalyst) and catalysis pyrolysis. They also designed by computer aided design (CAD) and machine for domestic purpose. This machine can be used for produce oil from waste plastics. Sachuthanathan and Rghurami Reddy [13] probed generation of diesel like fuel from city strong waste plastics for utilizing in CI Engine and furthermore study the burning, execution and discharge attributes of plastic oil. The performance and emission slightly increased with increase in load, increasing the other factors combustion efficiency increased. Boopathy Pradeep and Balakrishnan [8] investigated the conversion of plastic wastes into hydrocarbon fuels. The process includes following steps: collection of plastic waste, storing, shredding, feeding to hopper, PVC removing machine, flow into heating vessel in presence of catalyst and finally obtained liquid fuel such is petrol, diesel and kerosene. It is supporting a clean zero plastic waste environment. Anup and Vilas Watwe [3] concentrated on squander plastic pyrolysis oil as option for SI and CI motors with certain sorts of plastics as strong natural energizes, material in a round and hollow reactor at temperature of 300–350°C. Results express that petroleum motor ready to run with 100% waste plastic oil. Plastic pyrolysis oil exhibits higher thermal efficiency (50% for petrol and 75% for diesel engines) for the different load and produce less emissions of unburned hydrocarbons. Devy K Ratnasari [10] investigated the two stage pyrolysis catalysis of high energy density polyethylene for plastic pyrolysis. The first stage followed by catalysis of evolved hydro carbon pyrolysis gases in the second stage using solid acid catalysts produce gasoline range hydro carbon oil. Nabila Shehata and Hussain [9] analysed on fuel produced from catalytic pyrolysis of waste plastic raw material as PET of pepsi, coca cola bottles, LDPE of plastic bags, PS of yogurt plastic cups, PVC of oil bottles and temperature 400 °C for 5 h with catalyst natural zeolite. Miandad [7] despite all the potential advantages of catalytic pyrolysis. Some of the limitations such as high energy

demand, catalytic cost and less reuse of catalytic. Concludes this process solves problems of fuel shortage and pollution, it also represents low-cost production process to great support to Indian economy. Kaustav Mukharjee and Thamotharan [4], found that presentation and discharge trial of a few mixes of squanders plastic oil with diesel and ethanol on four stroke twin chamber diesel motor and delivered relatively higher motor execution. The further development of plastic pyrolysis process with catalyst may produce high quality fuel from plastic waste. Results plastic oil represents a fairly good alternative fuel and non hazardous.

3. Statement of problem

Most of Plastic waste winds up in land fill, ocean full and turn into a carbon sink, which it might take as long as 1000 years to deteriorate and spill toxins into soil and water. Plastic waste influences the people, creatures, fowls and earth and condition. About a fourth of all plastic waste is polypropylene used to make nourishment compartments, jugs and funnels.

4. Experimental details

4.1. Mini reactor

The plastic pyrolysis process is conducted in the mini reactor have 300 mm diameter and 600 mm height. It is made of stainless steel, the top of the reactor provided Hooper for charging plastic waste (Fig. 1).

4.2. Pistol thermometer

Digital non contact IR thermometer is used in this work. It is used to measure pyrolysis temperature from –50 °C to 680 °C with safety and quickly (Fig. 2).

Table 1 represents various levels and factors of Pyrolysis.

Taguchi design of experiment and L_9 orthogonal array is applied to conduct the experiments. Catalyst, temperature and production time are the input process parameters of production of fuel and response parameter is quantity of fuel.

4.3. Methodology

During conversion of plastic waste into Fuel follows the methodology (Fig. 3).



Fig. 1. Mini Reactor.



Fig. 2. Pistol Thermometer.

Table 1
Levels and factors of catalytic pyrolysis process.

Level	Catalyst (%)	Temperature (°C)	Production time (sec)
1	5	350	12
2	10	465	24
3	15	535	36

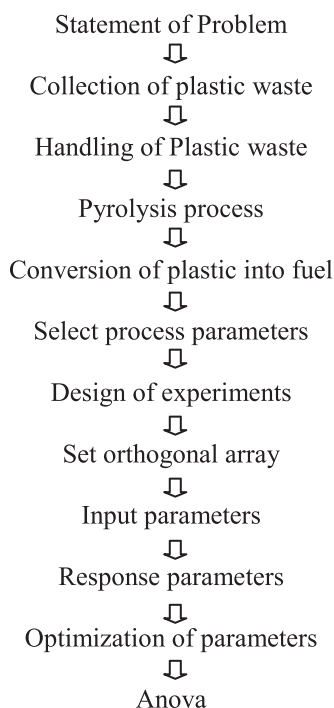


Fig. 3. Methodology.

Table 2
Catalytic pyrolysis process parameters.

Exp. No	Catalyst (%)	Temperature (°C)	Production time (sec)	Quantity of fuel (ml)	SNR
1	1	1	1	18	25.10
2	1	2	2	16	24.08
3	1	3	3	17	24.60
4	2	1	2	19	25.57
5	2	2	3	25	27.95
6	2	3	1	24	27.60
7	3	1	3	20	26.02
8	3	2	1	23	27.23
9	3	3	2	22	26.84

5. Result and discussion

In Catalysis pyrolysis process, the plastic waste shredded are charged into the hopper of Mini reactor and heated to 350 °C in the absence of oxygen. Here alumina is used as a catalyst. After 15 min charged shredded waste plastics are converted into gaseous fuel state and passed through condenser. The outlet of condenser is immersed in the Water bath and converted into liquid fuel. Result from plastic pyrolysis of LDPE using real time waste plastic in to oil form domestic and industrial sectors.

Table 2 denotes Catalytic pyrolysis process parameters for production fuel from plastic waste. It also represents larger quantity of fuel is produced by 2nd level (10%) catalyst, 2nd level of temperature (465 °C) and 3rd level of production time (36 sec) is best process parameters of Catalytic pyrolysis process for converting plastic waste into liquid fuel. It is graphically represented in main effect plot is shown Fig. 4.

Table 3 represent catalyst (Rank 1) is a dominating process parameter for production of fuel form waste plastic through plastic pyrolysis process. Optimize the liquid product of pyrolysis from as must of 5 kg of polypropylene (PP) plastic waste using a fixed bed type to minimize the oxygen entry the reactor.

Table 4 denotes influencing factor of catalytic pyrolysis process through higher value of 'F' (7.97) of source. It is also mentioned that most influencing process parameters of catalytic pyrolysis process is catalyst for conversion of plastic waste into liquid fuel. Ramli Thahir [14] found that liquid fuel obtained from pyrolysis process cannot be directly used as fuel due to presence of ash and wax from feed stock.

6. Conclusion

In this experimental work fuel is produced from plastic waste (polypropylene) using catalytic pyrolysis method. The following conclusions are occurs from this work.

1. Alternate fuel is delivered from plastic waste (polypropylene) effectively by synergist pyrolysis technique.
2. From this experiment 10% catalyst, 465 °C of temperature, 36 min of production time are the optimized process parameter for production of 25 g of oil from the plastic waste (polypropylene).
3. From Analysis of Variance, the higher value of 'F' denotes catalyst is an important and Influencing parameter for production of fuel from plastic waste (polypropylene).
4. From Interaction plot, catalyst, temperature and production time are the dependent parameter for production of fuel from plastic waste (polypropylene).
5. To further development of plastic pyrolysis process 25% of catalytic might be produce higher grade fuel for running automobile engines.

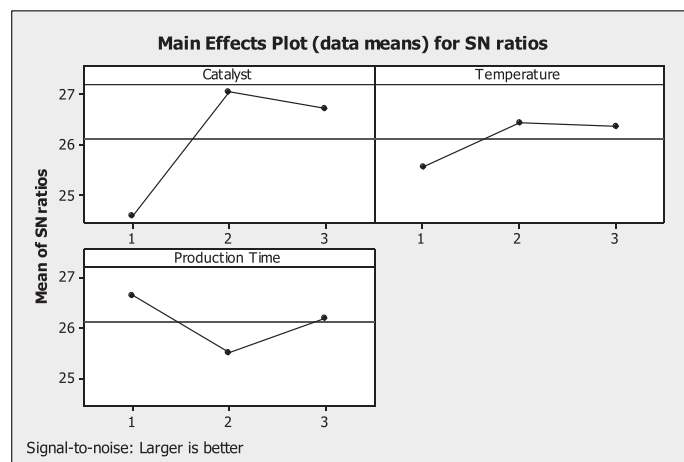


Fig. 4. Plot for pyrolysis process parameters production fuel from waste plastic.

Table 3

Response table for production of fuel from Plastic waste.

Levels	Catalyst (%)	Temperature (°C)	Production Time (Sec)
1.	24.60	25.57	26.65
2.	27.05	26.43	25.50
3.	26.70	26.35	26.20
Delta	2.45	0.86	1.15
Rank	1	3	2

Table 4

ANOVA Value of Production of fuel form Plastic waste.

Sources	DF	Seq SS	Adj SS	Adj MS	F	T
Catalyst	2	54.889	54.889	27.444	7.97	0.112
Temperature	2	9.556	9.556	4.778	1.39	0.419
Production Time	2	10.889	10.889	5.444	1.58	0.387
Error	2	6.889	6.889	3.444	–	–
Total	8	82.222	–	–	–	–

6. The study has been conducted to investigate the effects of temperature and optimize liquid products through condenser.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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